

Course Code	EC24P
Subject Professor In-charge	Prof. Satendra Mane
Student Name	Rahul Bhadane
Roll Number	23104B0003
Class	T.E – EXTC. SEM-V
Division	B
Date of Performance	1-8-25
Date of Submission	8-8-25

EXPERIMENT NO.3

Estimated time to complete this experiment: 02 hours.

Objective: To observe the Body effect for 180nm N-channel MOSFET.

CO to be achieved: CO1.

Expected Outcome of Experiment:

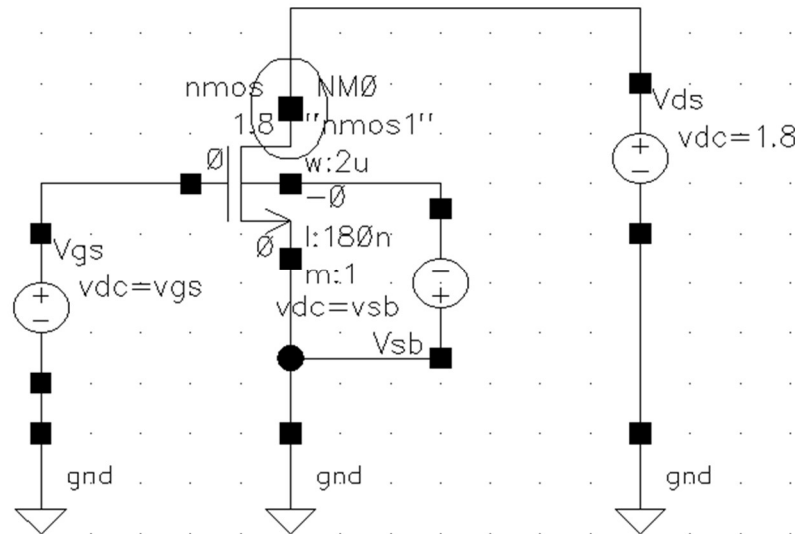
- Students will learn cadence virtuoso.
- Students will observe the effect of varying body bias voltage on transfer characteristics of MOSFET.

Pre Lab/ Prior Concept: Knowledge of working of MOSFET and Secondary effect in MOSFET.

Apparatus/TOOL:

- PC
- CADENCE VIRTUOSO.

Circuit Diagram:



5) Procedure.

- Open the cadence virtuoso tool in Red-Hat Linux. The commands are
 - `csh`
 - `source /home/install/cshrc`
 - `virtuoso.`
- Create the Library and cell view
- Open log window using terminal.
- Click on the new option, now create library with gpd180 from technology library.
- Again click the on new option, and set the cell view.
- Cadence Virtuoso dashboard will be opened.
- For characterization of 180nm NMOS Transistor
- Select NMOS transistor from add instance and select its channel length as 180nm (Nano meter) and channel width as W 1um (micro-meter).
- Construct the circuit shown in circuit diagram in Cadence.

- Take $V_{DD}=1.8V$ and $V_{GS}=1V$ and $V_{SB}=0V$.
- Then plot the Transfer Characteristics, for V_{SB} varying from 0 to 1.8V.
- From the transfer characteristics find the different values of Threshold voltage V_{TH} for different values of V_{SB} .
- Then calculate g_m of given NMOS Transistor for each value of width V_{SB} .

$$g_m = \frac{ID_2 - ID_1}{V_{GS2} - V_{GS1}} \quad \text{U}$$

- Then plot the output characteristics for different values of V_{SB} .
- From the output characteristics calculate r_o for each value of V_{SB} .

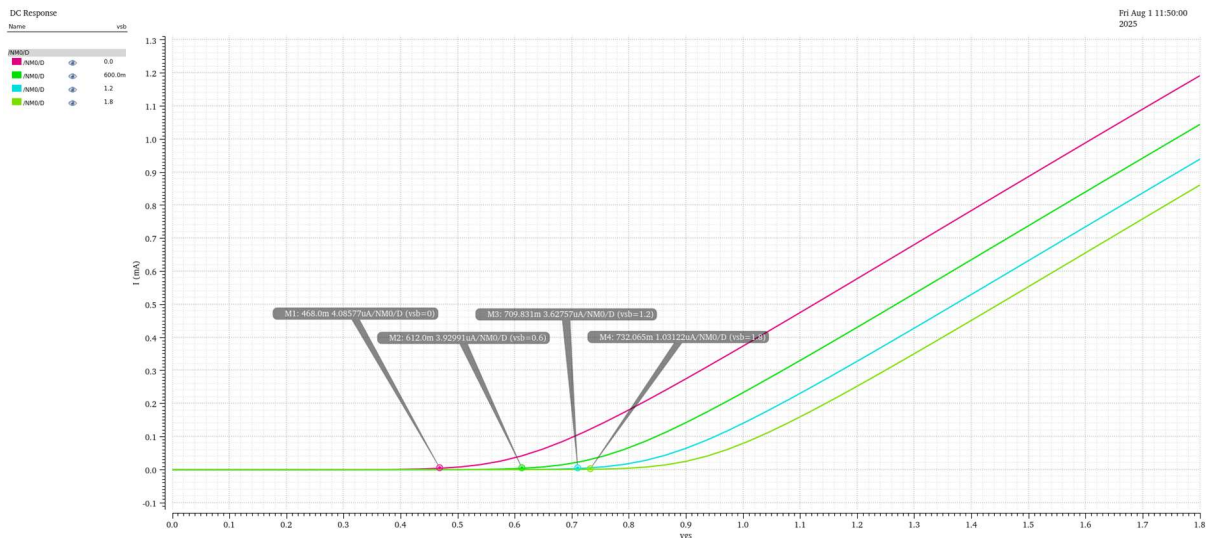
$$r_o = \frac{V_{DS2} - V_{DS1}}{ID_2 - ID_1} \Omega$$

- Then for given values of V_{SB} calculate the value of Body effect coefficient γ using the equation.

$$V_{TH} = V_{TH0} + \gamma \left(\sqrt{|2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|} \right) \quad .$$

$$\text{Where } V_{TH0}=0.5V \text{ and } |2\phi_F| = 0.7V$$

Results:



Observation Table :

Body bias voltage V_{SB}	V_{TH} (From graph).	γ (Body effect coefficient).
0V	0.5 V	0
0.6V	0.612 V	0.369
1.2V	0.735 V	0.433
1.8V	0.795 V	0.396

Calculation :

$V_{SB} = 0V$
$V_{TH} = 0.5V$
$V_{TH} = V_{TH0} + \gamma (\sqrt{ 2\phi_F + V_{SB}} - \sqrt{ 2\phi_F })$
$\gamma = 0$
$V_{SB} = 0.6V$
$V_{TH} = 0.612V$
$0.612 = 0.5 + \gamma (\sqrt{10.7 + 0.6} - \sqrt{10.7})$
$\gamma = 0.369$
$V_{SB} = 1.2V$
$0.735 = 0.5 + \gamma (\sqrt{10.7 + 1.2} - \sqrt{10.7})$
$\gamma = 0.433$
$V_{SB} = 1.8V$
$0.795 = 0.5 + \gamma (\sqrt{10.7 + 1.8} - \sqrt{10.7})$
$\gamma = 0.396$

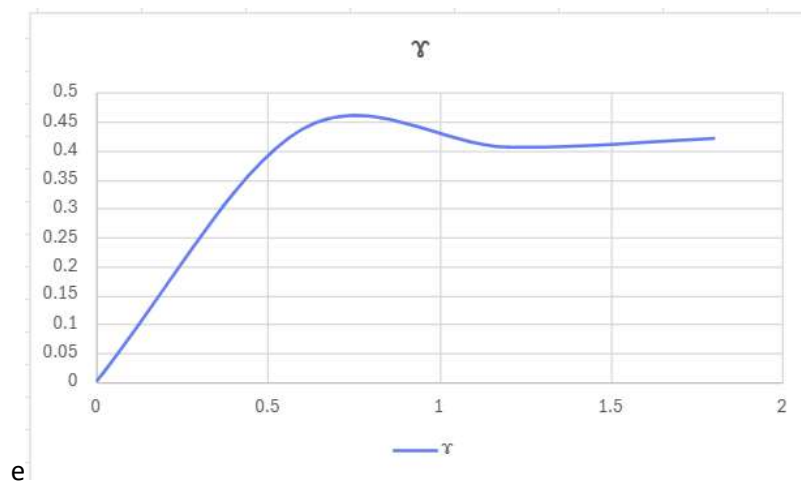
Conclusion :

In this experiment, the body effect in a 180 nm NMOS transistor was successfully observed using Cadence Virtuoso. By varying the body bias voltage V_{SB} , the threshold voltage V_{TH} was found to increase, confirming the influence of the body effect on the device characteristics. The calculated body effect coefficient γ showed that higher substrate bias leads to a noticeable change in the MOSFET's transfer and output characteristics. This experiment enhanced our understanding of secondary effects in MOSFETs and demonstrated how body biasing can affect device performance parameters like V_{TH} , g_m , I_{DSAT} , and intrinsic gain.

Post lab Questions :

Q) From the Observation table plot the following graph and for every graph write the comment.

- γ Vs V_{SB} .



Practical Performance (5 Marks)	Presentation and Conclusion (5 Marks)	Total (10 Marks)	Sign	Date of Submission.